Implementation of Fourth Order Chebyshev Bandpass Filter Using Folded Stepped-Impedance Resonator

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ABSTRACT

This poster presents the design and simulation of Chebyshev bandpass filter with folded stepped-impedance resonator aimed for the optimization of filter performance used for UHF applications. A 4th order Chebyshev bandpass filter was designed with a passband ripple of 0.0431 and a return loss of -20dB with a center frequency of 800MHz, fractional bandwidth of 6% and a characteristic impedance of 50Ω.The calculation and simulation result are compliant with each other.

INTRODUCTION

Filters are components designed with the purpose of selecting or transmitting signals over specified band and rejecting signals over the other bands. Although various forms of microwave filters are available in the literature, they all have the same common point by which their performance is evaluated using the distributed network concept, assuming that they usually consist of periodic structures exhibiting passband and stopband characteristics in various frequency bands [1].

The organization of this poster is as follows. In Section 3, a brief explanation on the proposed filter design is given and the investigation of the properties for the resonators of the filter is discussed in Section 3 as well.Finally, the results are discussed in Section 4 and conclusion is presented in Section 5.

PROCEDURE

In this section, a design and simulation of 4 pole Chebyshev bandpass filter is presented. The substrate that was utilized for the simulation was the RT /Duroid 6010LM with a thickness h=1.27mm and a dielectric constant ε_r=10.7mm. To design a 4 pole (4th order) Chebyshev bandpass filter, it was assumed that the circuit was an ideal lossless condition. Firstly, the standard 4th order normalized low pass filter was transformed into bandpass filter with the series and parallel LC circuit as shown in Fig. 1. The bandpass filter introduced J-inverters (Fig. 2) and these J-inverters were replaced with identical LC networks as shown in Fig. 3. Secondly, a folded stepped-impedance resonator was designed by approximating the values by multiple rounds of simulation in the ADS software to resonate at a centre frequency of 800 MHz, as shown in Fig. 4, based on the calculations given below: w $=\frac{8he^{A}}{e^{2A}-2}$, A $=\frac{Z0}{60}\sqrt{\frac{\varepsilon_r+1}{2}}+\frac{\varepsilon_r-1}{\varepsilon_r+1}\left[0.23+\frac{0.11}{\varepsilon_r}\right]$, $\varepsilon_{eff}=\frac{\varepsilon_r+1}{2}+\frac{\varepsilon_r-1}{2}\left(1+\frac{12h}{w}\right)^{-0.5}$, l$=\frac{\lambda_g}{2}=\frac{C_0}{2fo\sqrt{\varepsilon_{eff}}}$, $K_{i,i+1}=\frac{FBW}{\sqrt{g_1g_2}}$, $Q_{ext}=\frac{g_0g_1}{FBW}$ [2], where $K_{i,i+1}$=coupling distance between every two resonators, Q_{ext}=external quality factor, λ_g is the guided wavelength and l and w are the length of the resonator and the width of the ports respectively. The coupling factor between every two resonators and the external quality factor of the resonators were simulated and their practical values were calculated using these two equations $Q_{ext}=\frac{f_0}{f_2-f_1}$, $K_{i,i+1}=\frac{f_2^{\,2}-f_1^{\,2}}{f_2^{\,2}+f_1^{\,2}}$ [2] then compared (Fig. 8) with the theorical values calculated from the above equations. Finally, a combined microstrip folded stepped-impedance resonator with all 4 identical resonators was designed by again approximating the values to get the desired result as shown in Fig. 5.

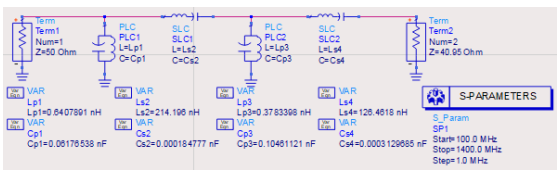

Fig. 1: circuit model with different series and parallel LC lumped circuit elements

L_{sk}, C_{sk}, L_{pk} and C_{pk}[2] calculations with k=1,...,4.

$C_{pk}=\frac{g_k}{Z_0\omega_0 FBW}$, $L_{pk}=\frac{Z_0\,FBW}{g_k\omega_0}$

$C_{sk}=\frac{FBW}{g_k\omega_0 Z_0}$, $L_{sk}=\frac{g_kZ_0}{\omega_0\,FBW}$

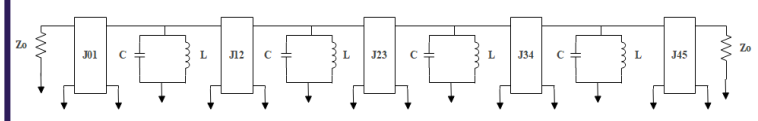

Fig. 2: circuit model with identical LC resonators and J-inverters

J_{01} and $J_{k,k+1}$ [2] calculations with k=1,...,4. $J_{01}=\frac{g_0}{Z_0}$, $J_{k,k+1}=\sqrt{\frac{g_1^{\,2}}{g_kg_{k+1}}}/Z_0$

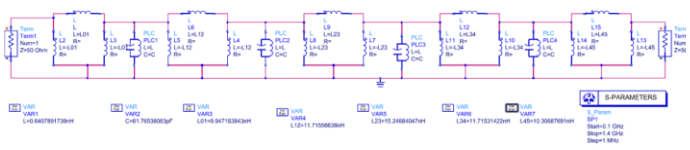

Fig. 3: Final circuit model with all J-inverters replaced by inductors only

J-inverters inductors values calculations $L_{Jk.k+1}$ with k=0,...,4. $L_{Jk.k+1}=\frac{1}{J_{k,k+1}2\pi f_0}$ [2]

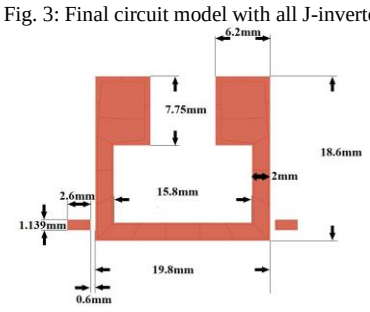

Fig. 4: Stepped folded-impedance resonator

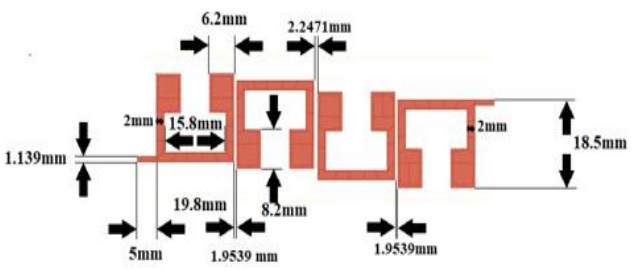

Fig. 5: Microstrip folded stepped-impedance resonator implementation of the filter

FINDINGS

The circuit model of different LC lumped circuit elements and the circuit model of identical LC resonators and J-inverters results are shown in Fig. 6. The microstrip of the 4th order Chebyshev bandpass filter results was presented in Fig. 7 below. A table (Fig. 8) shows a comparison between practical (simulated) and theorical (calculated) values.

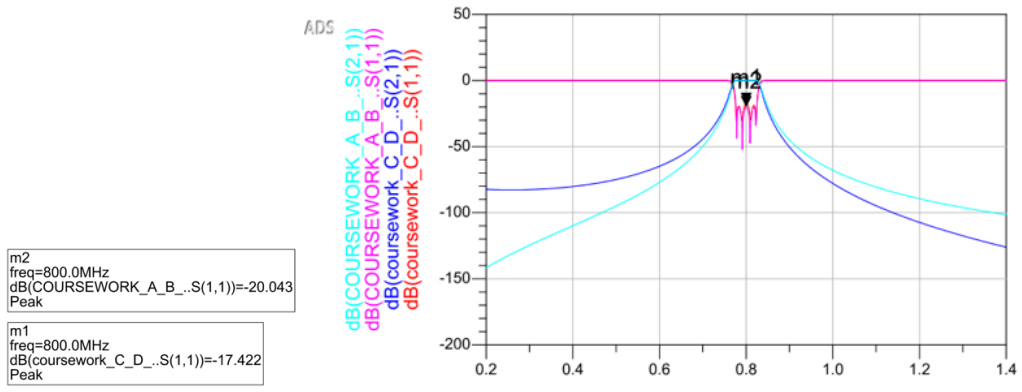

Fig. 6: Comparison between result of circuit model of different LC lumped circuit elements and identical LC resonators and J-inverters

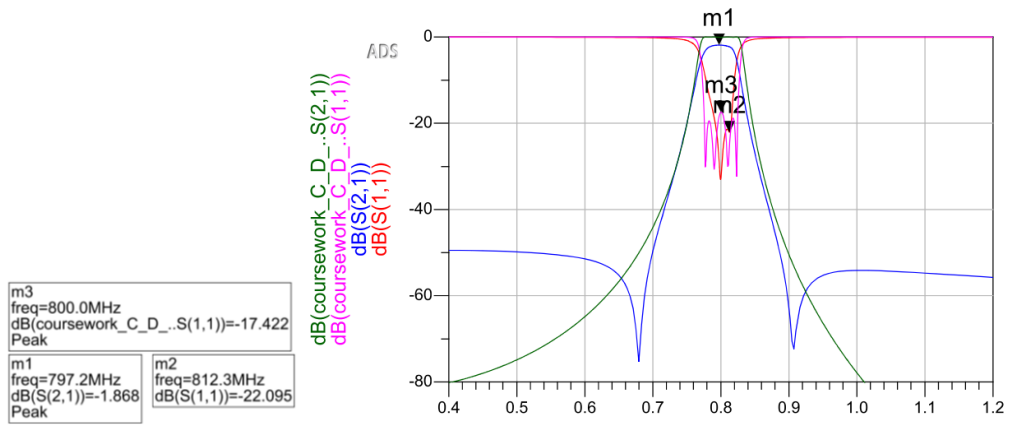

Fig. 7: Comparison of results for microstrip with identical LC resonators and J-inverters

PARAMETERS	l (mm)	K_{12}	K_{23}	K_{34}	Q_{ext}
Practical	72.5	0.0547	0.0420	0.0547	11.13
Theorical	70.22	0.0547	0.0420	0.05469	15.52

Fig. 8: Table for comparing theorical and practical values

From the graphs above, in Fig. 6, the circuit model of different LC lumped circuit elements (sky-blue line) and the circuit model of identical LC circuit with all J-inverters replaced by inductors only network design (dark blue line) showed that the circuit was in an ideal lossless condition having 0 insertion loss. Moreover, from Fig. 6, the return loss for the circuit model of different LC lumped circuit elements (pink line) showed a return loss better than -20dB whereas for the circuit model of identical LC resonators and all J-inverters replaced by inductors only network design (red line) showed a mere return loss of -17.422dB. In Fig.7, the circuit model of identical LC circuit with all J-inverters replaced by inductors only network design (Fig. 3) represented an ideal condition (pink line) Whereas the microstrip resonators depicted a practical condition of use (red line). The microstrip resonators produced both dielectric and conductor losses. From Fig.7, the microstrip resonators showed minimum insertion loss better than -1.87dB (dark blue line) and a return loss was better than -22 dB (red line). The lumped circuit with identical LC resonators and all J-inverters replaced by inductors only network design graph (green line on Fig.7) on the other hand showed 0 insertion loss.

CONCLUSION

A 4 pole Chebyshev bandpass filter has been designed and simulated using folded stepped-impedance resonator. The circuit model and microstrip layout showed good agreement at the required centre frequency of 800 MHz. The simulation results of return loss better than -22dB and insertion loss better than -1.87dB were achieved at a centre frequency of 797.2 MHz closer to the required centre frequency of 800 MHz. Further research is recommended to achieve the most precise values for an ideal microstrip layout of the 4 pole Chebyshev bandpass filter.

REFERENCES

[1] Pozar, D.M. Microwave Engineering; Wiley: Hoboken, NJ, USA, 2012

[2] A. O. Nwajana, "Circuit modelling of bandpass/channel filter with microstrip implementation," *Indonesian Journal of Electrical Engineering and Informatics (IJEEI)*, vol. 8, no. 4, 2020.